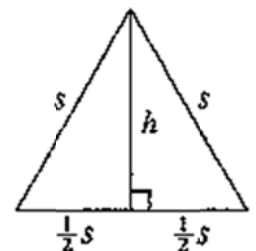
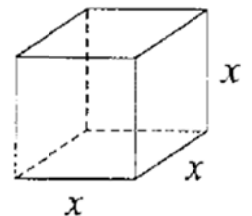
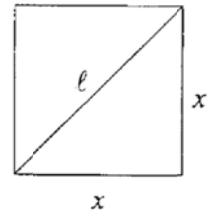
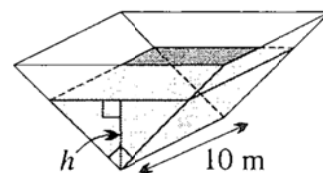


Rates of Change

- Given that $y = x^3 + x$, differentiate with respect to time, using the chain rule.
 - If $\frac{dx}{dt} = 5$, find $\frac{dy}{dt}$ when $x = 2$.
 - If $\frac{dy}{dt} = -6$, find $\frac{dx}{dt}$ when $x = -3$.
- A circular oil stain of radius r and area A is spreading on water. Differentiate the area formula with respect to time to show that $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$. Hence find:
 - The rate of increase of area when $r = 40 \text{ cm}$ if the radius is increasing at 3 cm/s .
 - The rate of increase of the radius when $r = 60 \text{ cm}$ if the area is increasing at $10 \text{ cm}^2/\text{s}$.
- A spherical bubble of radius r is shrinking so that its volume V is decreasing at a constant rate of $200 \text{ cm}^3/\text{s}$. Show that $\frac{dV}{dt} = 4\pi r^3 \frac{dr}{dt}$.
 - At what rate is its radius decreasing when the radius is 5 cm ?
 - What is the radius when the radius is decreasing at $2\pi \text{ cm/s}$?
 - At what rate is the radius decreasing when the volume is $36\pi \text{ cm}^3$? [Hint: first find the radius when the volume is $36\pi \text{ cm}^3$.]
- The side length x of a square shadow is increasing at 6 cm/s . If the area is A and the length of the diagonal is ℓ , show that $\frac{dA}{dt} = 2x \frac{dx}{dt}$ and $\frac{d\ell}{dt} = \sqrt{2} \frac{dx}{dt}$. Hence find the rates of increase of the area and the diagonal when
 - The side length is 70 cm ,
 - The area is 1 m^2 .
- The side length x of a shrinking cube is decreasing at a constant rate of 5 mm per minute. Show that the rates of change of volume V , surface area A , and the total edge length ℓ are $\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$ and $\frac{dA}{dt} = 12x \frac{dx}{dt}$ and $\frac{d\ell}{dt} = 12 \frac{dx}{dt}$.
 - Find the rate at which volume, surface area and edge length are decreasing when:
 - The side length is 30 cm ,
 - The volume is 8000 cm^3 .
 - Find the side length when the volume is decreasing at $300 \text{ cm}^3/\text{min}$.
- Show that in an equilateral triangle of side length s , the area is given by $A = \frac{1}{4}s^2\sqrt{3}$ and the height by $h = \frac{1}{2}s\sqrt{3}$.
 - Find formulae for the rates of change of area and height.
 - Hence find the rate at which the area and the height are increasing when the side length is 12 cm and is increasing at 3 mm/s .
- Water is being poured into an inverted right conical vessel whose apex angle is 90° at a constant rate of $3 \text{ cm}^3/\text{s}$. At what rate is the water level rising when the depth is $\pi \text{ cm}$?
- The perimeter of a circular sector is 20 cm . The radius is increasing at the rate of 5 cm/s . At what rate is the angle of the sector changing when the radius length is 10 cm ? Also at what rate is the area changing when the radius length is 10 cm ?
- A ladder 5 metres long has its upper end against a vertical wall and its lower end on the horizontal floor. The lower end moves away from the wall at a constant speed of 1 m/s . Find the speed at which the upper end moves down the wall 4 s after the lower end has left the wall.



10. The altitude of a right-angled triangle is 6 cm and the base is increasing at a constant rate of 2 cm/s. At what rate is the hypotenuse increasing when its length is 10 cm?
11. A loading chute is in the shape of a right square pyramid of base length 10 m and depth 8 m. Liquid is poured in the top at a rate of $4\text{ m}^3/\text{min}$. At what rate is the level rising when the depth is 4 m?
12. A kite, 50 metres high, is being carried horizontally by the wind at a rate of 4m/sec. How fast is the string being let out, when the length of the string is 100 metres?
13. A spherical balloon is to be filled with water so that its surface area increases at a constant rate of $1\text{ cm}^2/\text{s}$.
- Find, when the radius is 3 cm:
 - The required rate of increase of the radius.
 - The rate at which the water must be flowing in at that time.
 - Find the volume when the volume is increasing at $10\text{ cm}^3/\text{s}$.
14. A water trough is 10 metres long, with cross section a right isosceles triangle. Show that when the water has depth $h\text{ cm}$, its volume is $V = 1000h^2$ and its surface area is $A = 2000h$.
- Find the rates at which depth and surface area are increasing when the depth is 60 cm if the trough is filling at 5 litres per minute (remember that 1 litre is 1000 cm^3).
 - Find the rates at which the volume and the surface area must increase when the depth is 40 cm, if the depth is required to increase at a constant rate of $0.1\text{ cm}/\text{min}$.
15. The equation of the path of a bullet fired into the air is $y = -20x(x - 20)$, where x and y are displacements in metres horizontally and vertically from the origin. The bullet is moving horizontally at a constant rate of $\frac{1}{2}\text{ m}/\text{s}$.
- Find the rate at which the bullet is rising:
 - When $x = 8$,
 - When $x = 18$,
 - When $x = 10$,
 - When $y = 1500$.
 - Find the height when the bullet is:
 - Rising at $30\text{ m}/\text{s}$,
 - Falling at $70\text{ m}/\text{s}$.
 - Use the gradient function $\frac{dy}{dx}$ to find the angle of flight when the bullet is rising at $10\text{ m}/\text{s}$.
 - How high does the bullet go, and how far away does it land?
16. Sand being poured from a conveyor belt forms a cone with height h and semi-vertical angle 60° . Show that the volume of the pile is $V = \pi h^3$, and differentiate with respect to t .
- Suppose that the sand is being poured at a constant rate of $0.3\text{ m}^3/\text{min}$, and let A be the area of the base. Find the rate at which the height is increasing:
 - When the height is 4 metres,
 - When the radius is 4 metres.
 - Show that $\frac{dA}{dt} = 6\pi h \frac{dh}{dt}$, and find the rate of increase of the base area at these times.
 - At what rate must the sand be poured if its required that the height increase at $8\text{ cm}/\text{min}$, when the height is 4 metres?



17. An upturned cone of semi-vertical angle 45° is being filled with water at a constant rate of $20 \text{ cm}^3/\text{s}$. Find the rate at which the height, the area of the water surface, and the area of the cone wetted by the water, are increasing when the height is 50 cm .
18. A square pyramid has height twice its side length s .
- Show that the volume V and the surface area A are $V = \frac{2}{3}s^3$ and $A = (\sqrt{17} + 1)s^2$.
 - Hence find the rate at which V and A are decreasing when the side length is 4 metres if the side length is shrinking at 3 mm/s .
19. A ladder 13 metres long rests against a wall, with its base x metres from the wall and its top y metres high. Explain why $x^2 + y^2 = 169$, solve for y , and differentiate with respect to t . Hence find, when the base is 5 metres from the wall:
- The rate at which the top is slipping down when the base is slipping out at 1 cm/s ,
 - The rate at which the base is slipping out when the top is slipping down at 5 mm/s .
20. Water is flowing into a hemispherical container of radius 10 cm at a constant rate of 6 cm^3 per second. It is known that the formula for the volume of a solid segment cut off a sphere is $V = \frac{\pi}{3}h^2(3r - h)$, where r is the radius of the sphere and h is the height of the segment.
- Find the rate at which the height of the water is rising when the water height is 2 cm .
 - Use Pythagoras' theorem to find the radius of the circular water surface when the height is h , and hence find the rate of increase of the surface area when the water height is 2 cm .

Answer

- $\frac{dy}{dt} = (3x^2 + 1)\frac{dx}{dt}$; **a.** 65; **b.** $-\frac{3}{14}$.
- a.** $240\pi \text{ cm}^2/\text{s}$; **b.** $\frac{1}{12\pi} \text{ cm/s}$.
- a.** $\frac{2}{\pi} \text{ cm/s}$; **b.** $\frac{5}{\pi} \text{ cm}$; **c.** $\frac{50}{9\pi} \text{ cm/s}$
- a.** $840 \text{ cm}^2/\text{s}$, $6\sqrt{2} \text{ cm/s}$; **b.** $1200 \text{ cm}^2/\text{s}$, $6\sqrt{2} \text{ cm/s}$
- a. i.** $1350 \text{ cm}^3/\text{min}$, $180 \text{ cm}^2/\text{min}$, 6 cm/min , **ii.** $600 \text{ cm}^3/\text{min}$, $120 \text{ cm}^2/\text{min}$, 6 cm/min ; **b.** $10\sqrt{2} \text{ cm}$
- a.** $\frac{dA}{dt} = \frac{1}{2}s\sqrt{3}\frac{ds}{dt}$, $\frac{dh}{dt} = \frac{1}{2}\sqrt{3}\frac{ds}{dt}$; **b.** $\frac{9}{5}\sqrt{3} \text{ cm}^2/\text{s}$, $\frac{3}{20}\sqrt{3} \text{ cm/s}$
- $\frac{3}{\pi^3} \text{ cm/s}$
- $-1^\circ/\text{s}$; $-50 \text{ cm}^2/\text{s}$
- $1\frac{1}{3} \text{ m/s}$
- 1.6 cm/s
- $\frac{4}{25} \text{ m/min}$
- $2\sqrt{3} \text{ m/s}$
- a. i.** $\frac{1}{24\pi} \text{ cm/s}$, **ii.** $1\frac{1}{2} \text{ cm}^3/\text{s}$; **b.** $\frac{32000\pi}{3} \text{ cm}^3$
- a.** $\frac{1}{24} \text{ cm/min}$, $83\frac{1}{3} \text{ cm}^2/\text{min}$; **b.** 8 litre/min , $200 \text{ cm}^2/\text{min}$
- a. i.** 40 m/s , **ii.** -160 m/s , **iii.** Travelling horizontally, **iv.** 100 m/s or -100 m/s ; **b. i.** 1955 metres, **ii.** 1755 metres; **c.** $\frac{dy}{dx} = 20$, angle is about $87^\circ 8'$; **d.** 2 km high, 20 metres away
- $\frac{dV}{dt} = 3\pi h^2 \frac{dh}{dt}$; **a. i.** $\frac{1}{160\pi} \text{ m/min}$, **ii.** $\frac{3}{160\pi} \text{ m/min}$; **b.** $\frac{dA}{dt} = 6\pi h \frac{dh}{dt}$, $\frac{3}{20} \text{ m}^2/\text{min}$, $\frac{3}{20}\sqrt{3} \text{ m}^2/\text{min}$; **c.** $\frac{96\pi}{25} \text{ m}^3/\text{min}$
- $\frac{1}{125\pi} \text{ cm/s}$, $\frac{4}{5} \text{ cm}^2/\text{s}$, $\frac{4}{5}\sqrt{2} \text{ cm}^2/\text{s}$
- b.** $0.096 \text{ m}^3/\text{s}$, $\frac{3}{125}(\sqrt{17} + 1) \text{ m}^2/\text{s}$
- $\frac{dy}{dt} = -\frac{x}{\sqrt{169-x^2}}\frac{dx}{dt}$; **a.** $\frac{5}{12} \text{ cm/s}$; **b.** $1\frac{1}{5} \text{ cm/s}$
- a.** $\frac{dV}{dt} = \pi h(20 - h)\frac{dh}{dt}$, $\frac{1}{6\pi} \text{ cm/s}$; **b.** radius = $\sqrt{2hr - h^2}$, $\frac{8}{3} \text{ cm}^2/\text{s}$